

3 (a)

$$F = qE = q \frac{\Delta V}{d}$$

$$1.12 \times 10^{-16} = 1.6 \times 10^{-19} \frac{\Delta V}{0.10}$$

[C1]

$$\Delta V = V_A - 0 = 70.0 \text{ V}$$

$$V_A = 70.0 \text{ V}$$

[A1]

(b)(i) Gain in PE = Loss in K.E. of electron

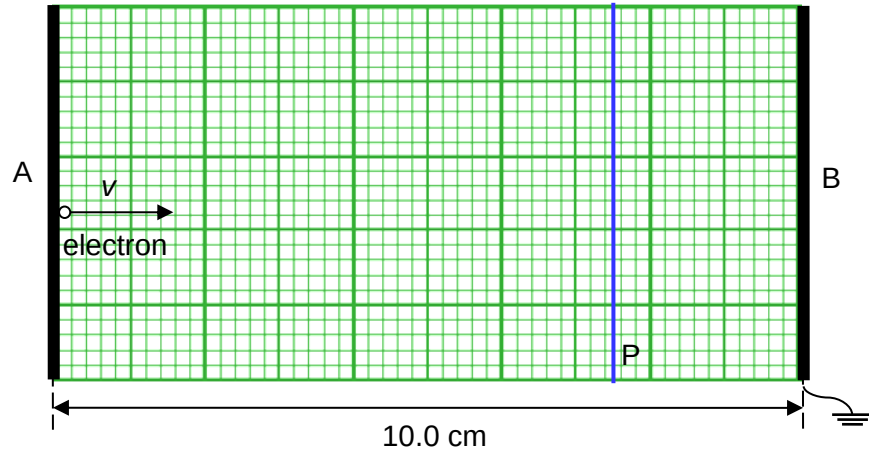
$$q\Delta V = \frac{1}{2}mv^2 \quad [C1]$$

$$-1.6 \times 10^{-19} \Delta V = \frac{1}{2} \times 9.11 \times 10^{-31} \times (4.30 \times 10^6)^2$$

$$V_P - 70 = -52.6 \text{ V} \quad [A1] \text{ (ecf)}$$

$$V_P = 17.4 \text{ V}$$

(b)(ii)



(b)(ii) $E = \Delta V/d = 70/10 = 7.0 \text{ V cm}^{-1}$

$\therefore$ for $\Delta V = 52.6 \text{ V}$, $d = 52.6/7.0 = 7.5 \text{ cm}$ [C1] (ecf)

Draw a vertical line 7.5 cm from plate A [M1]

(c) The electron will stop **before** the equipotential line as the horizontal component of velocity in the direction of the field would be lower. [A1]

(d)(i) $F = ma = qE$

$$1.67 \times 10^{-27} a_y = 1.6 \times 10^{-19} \times 2.0 \times 10^4$$

$$a_y = 1.92 \times 10^{12} \text{ m s}^{-2}$$

$$s_y = \frac{1}{2} a_y t^2$$

$$\frac{1}{2} (0.50 \times 10^{-2}) = \frac{1}{2} (1.92 \times 10^{12}) t^2$$

$$t = 5.1 \times 10^{-8} \text{ s} \quad [C2]$$

$$s_x = u_x t = 3.5 \times 10^6 \times 5.1 \times 10^{-8} = 0.179 \text{ m} \quad [A1]$$

(d)(ii) $a_{\text{proton}} = qE / m$

$$a_{\text{alpha}} = 2qE / 4m = \frac{1}{2} a_{\text{proton}} = 9.6 \times 10^{11} \text{ m s}^{-2} \quad [C1]$$

Same vertical displacement for both proton and alpha

[C1]

$$t_{\alpha} = 7.2 \times 10^{-8} \text{ s}$$

Using $s_x = u_x t$, since u_x is the same for both, $(s_x)_{\alpha} = 0.252 \text{ m}$

therefore alpha particle will exit the plates [A1]

4. (a)

The heating effect is due to power dissipation in the coil which is dependent on